

Office PDF Capability Sheet

Cross-platform PDF creation, inspection, extraction, form filling, and rendered visual QA.

What this demonstrates

This searchable PDF was generated with a pure Python pipeline. The same helper can inspect an existing PDF, render pages to PNG for visual QA, extract text, merge files, and fill form fields while preserving the source file.

Validation is part of the workflow, not an afterthought: every layout-sensitive PDF should be rendered to images and reviewed for clipping, missing glyphs, blank pages, and table overflow.

Quality gate: inspect the PDF structure, render the pages, then report both the final PDF and the preview artifacts.

Capability matrix

Route	Purpose	Default tool
CREATE	Generate polished searchable PDFs	ReportLab
INSPECT	Read pages, metadata, fields, text counts	pypdf
RENDER	Create PNG previews for visual QA	PyMuPDF
EXTRACT	Review text content	pdfplumber / pypdf
FILL	Write form field values	pypdf
MERGE	Combine source PDFs safely	pypdf

Rendered QA coverage

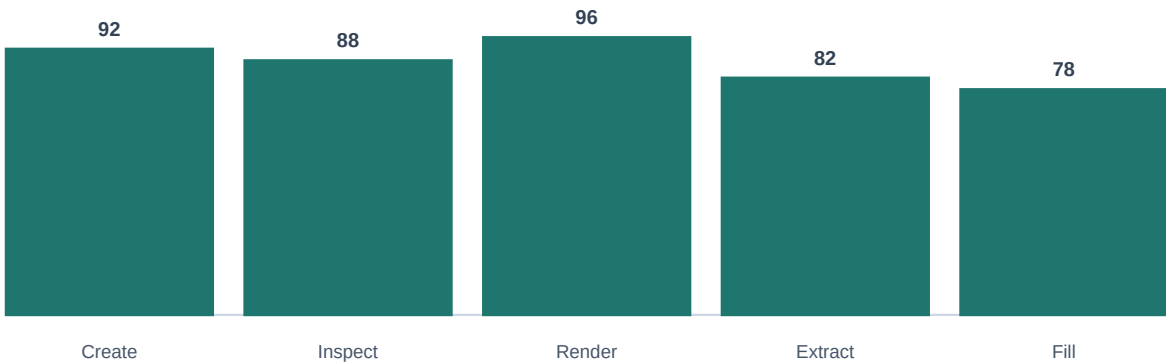


Figure 1: Example capability coverage shown as a generated vector chart.

Practical rule: when a PDF is important enough to deliver, it is important enough to render and inspect. This is the main difference between a casual PDF script and a reusable Office PDF skill.